**Situation 1** Manimala prepares vegetable soup for her family of 4 members. She uses 2 cup of vegetables, 600 ml of water, 1 teaspoon of salt and $\frac{1}{2}$ teaspoon of pepper. Suddenly her aunt and uncle join them. What would be the change in quantities of the ingredients to prepare soup for all the six members?

**Situation 2** In a military camp there are 200 soldiers. Grocery for them are available for 40 days. If 50 more soldiers join them, how long will these grocery last?

In the above situations a change in one quantity results in the corresponding change in other quantity. That is in situation 1, more the number of members, more will be the quantity of items. In situation 2, the more number of soldiers in the group, the grocery needed will be more and so it will last for less number of days.

In these situations when one quantity varies that brings a change in other quantity in two ways, that is,

(i) both the quantities increase or decrease.

(ii) increase in one quantity causes decrease in the other quantity or vice-versa.

If such quantities varies in constant ratio, they are said to be in proportion. Two different variations give rise to two types of proportion which will be discussed now.

**MATHEMATICS ALIVE-DIRECT AND INVERSE PROPORTION IN REAL LIFE**



Direct proportion




Inverse proportion

## 4.2 Direct Proportion

For example, if the cost of a shirt is ₹ 500, then the price of 3 shirts will be ₹ 1,500. The price of the shirts increases as the number of shirts increases. Proceeding the same way we can find the cost of any number of such shirts.

When observing the above situation, two quantities namely the number of shirts and their prices are related to each other. When the number of shirts increases, the price also increases in such a way that their ratio remains constant.

Let us denote the number of shirts as $x$ and the price of shirts as $y$ rupees. Now observe the following table

| Number of shirts ($x$) | 1 | 2 | 3 | 6 | 7 | ... |
|---|---|---|---|---|---|---|
| Price of shirts in ₹ ($y$) | 500 | 1000 | 1500 | 3000 | 3500 | ... |

From the table, we can observe that when the values of $x$ increases the corresponding values of $y$ also increases in such way that the ratio $\frac{x}{y}$ in each case has the same value



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